

DDR基础课程

第八节 - DDR控制器基本结构

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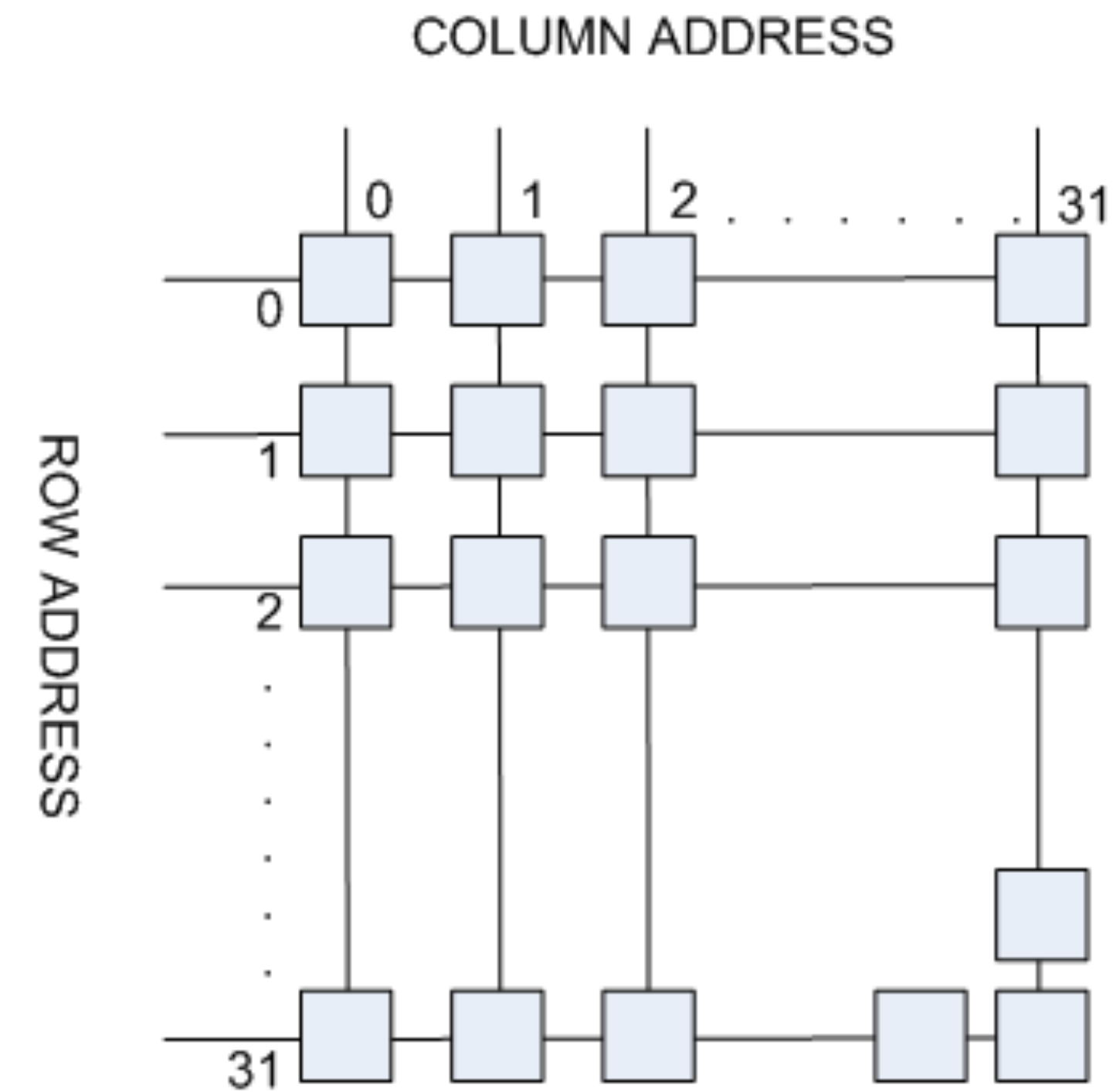
目录

关注的问题

- DDR控制器的主要功能目标?
- 典型常见特性?
- 主要架构划分?
- DFI接口基本介绍?

主要功能目标

- 功能
 - 在AXI协议与DDR协议之间进行转换
 - 保证DRAM的存储数据完整性
- 性能
 - 保证效率尽可能高
 - 保证延迟尽可能小



常见典型特性

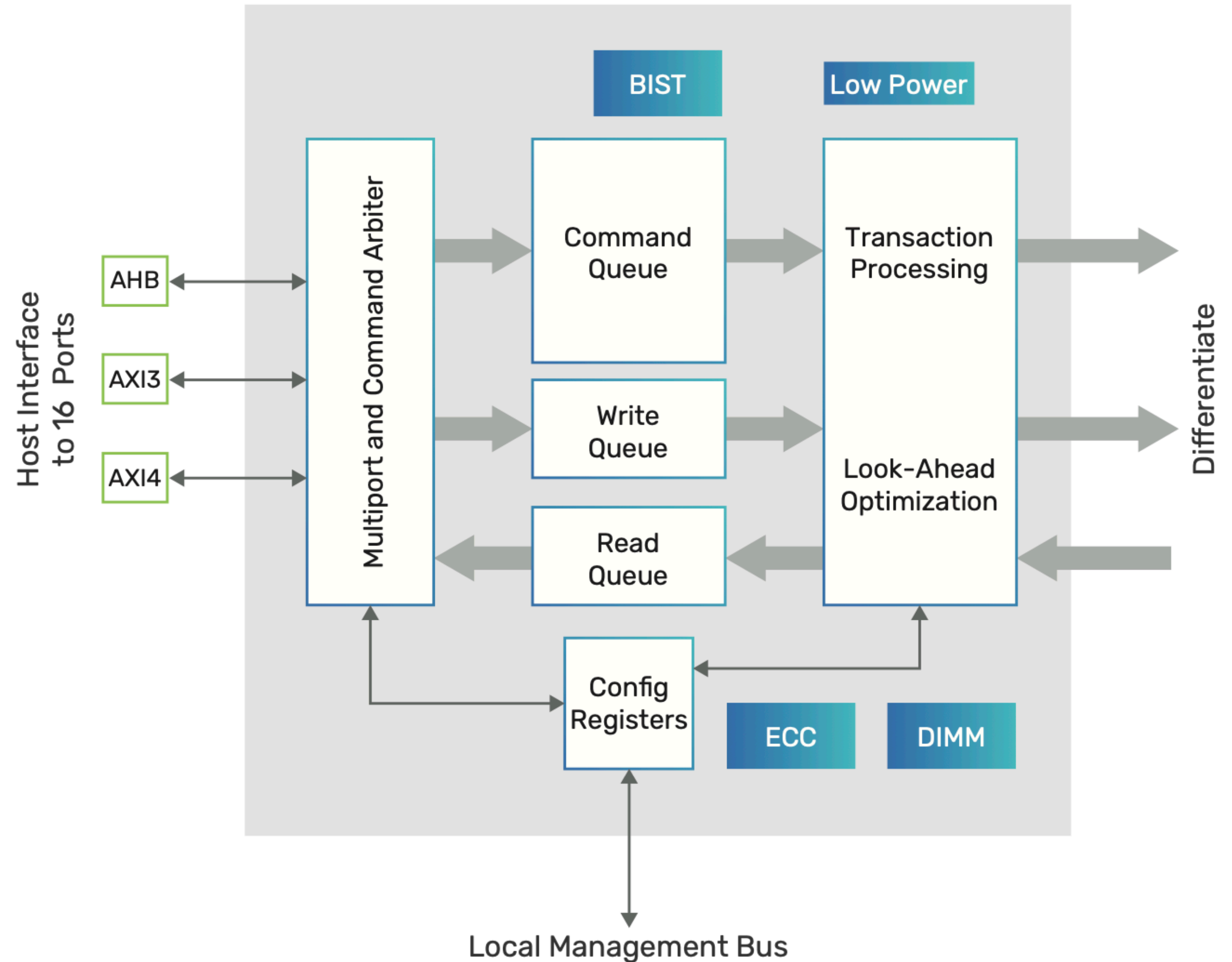
- ECC - Error Correction Code
- RAS - Reliability, Accessibility, Serviceability
- QoS - Quality of Service
- ODT - On-Die Termination

Key Features

- | | |
|---|---|
| ▶ Compliant to LPDDR5/4X4/3 and DDR5/4/3L/3 protocol memories | ▶ Supports advanced RAS features including SEC/DED ECC, error scrubbing, parity, etc. |
| ▶ Side-band or in-line ECC | ▶ Priority-per command on Arm® AMBA® 3 AXI and low-latency Denali interface |
| ▶ Single and multi-port host interface options | ▶ QoS features allow command prioritization on Arm AMBA 4 AXI interfaces |
| ▶ Flexible paging policy including auto-precharge-per-command | ▶ Silicon proven and shipping in volume |
- Maximizes bus efficiency via look-ahead command processing, bank management, auto-precharge and additive latency support
 - Latency minimized via parameterized pipelining
 - Achieves high clock rates with minimal routing constraints
 - Supports full-rate and half-rate clock operation
 - Multi-mode controller support
 - Full run-time configurable timing parameters and memory settings
 - Supports ODT, dynamic ODT, 2T timing and write leveling calibration
 - DFI compatible
 - Full set of Add-On cores available
 - Minimal ASIC gate count
 - Broad range of ASIC and FPGA platforms supported
 - Delivered fully integrated and verified with target PHY

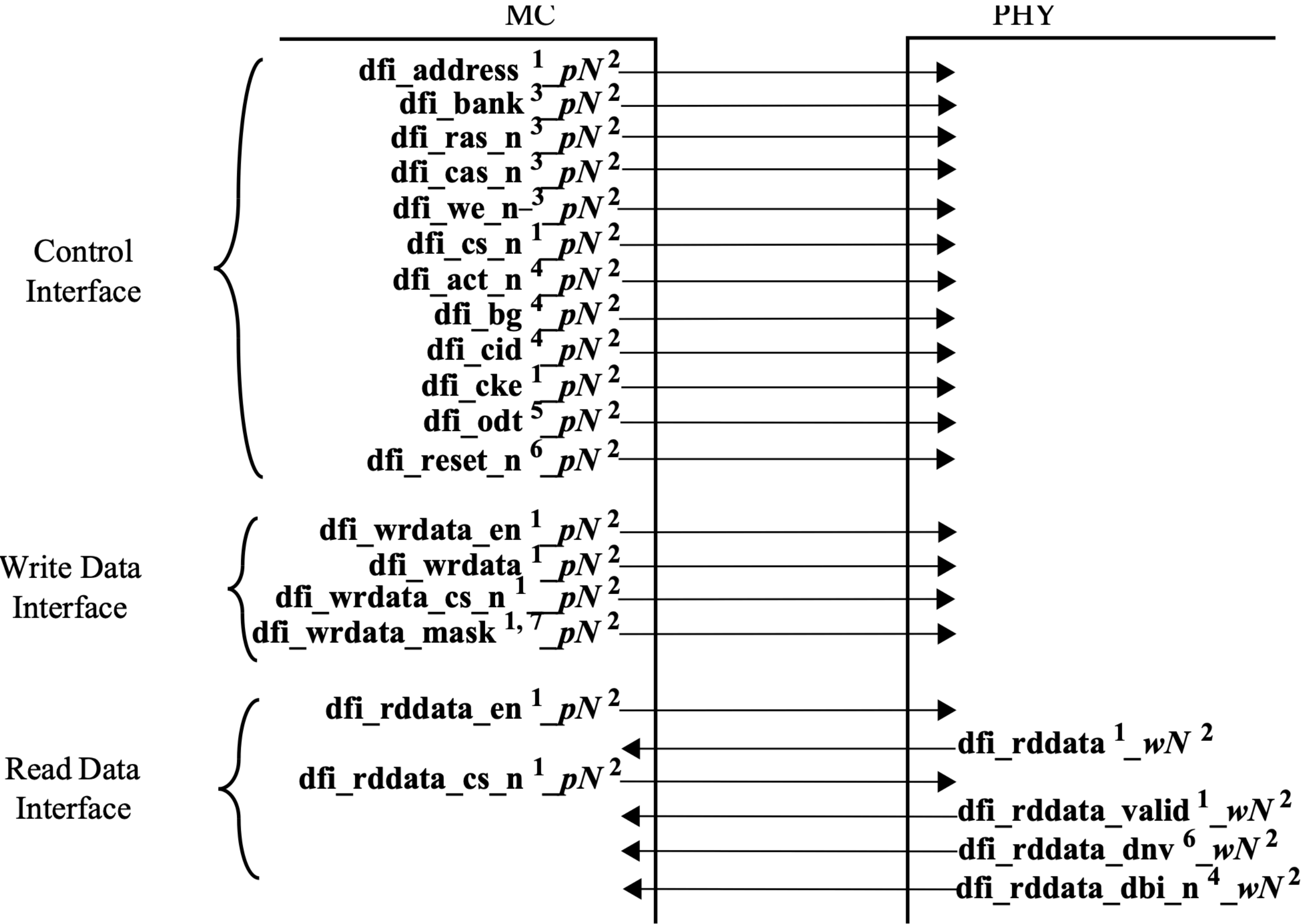
基本结构

- Address Mapping
- Out of Order Optimization
- Read Write Switch



DFI接口基础介绍

Interface Group	Description
Control	Required to drive the address, command, and control signals to the DRAM devices.
Write Data	Used to send write and receive valid read data across the DFI.
Read Data	
Update	Provides an ability for the MC or the PHY to interrupt and idle the DFI.
Status	Used for system initialization, feature support, and to control the presence of valid clocks to the DRAM interface.
Training	Used to execute gate training, read data eye training, and write leveling operations.
Low Power Control	Allows the PHY to enter power-saving modes.
Error	Used to communicate error information from the PHY to the MC.



上节课后问题解答

- 课后问题1: “DDR不同于CPU, GPU等核心计算单元, 对于SoC的作用是辅助性的”。以上说法是正确的吗?
错误。DDR作为整个计算环节的关键路径之一, 对于整个SoC的性能起到至关重要的作用。
- 课后问题2: “DDR因为其对于SoC的关键作用, 其物理摆放位置通常也位于SoC的中心位置”。以上说法是正确的吗? 错误。DDR虽然起到了关键作用, 但因为其需要与片外DRAM进行互联, 所以在floorplan上通常位于边缘。
- 课后问题3: “DDR的组成部分通常包括DDR控制器和物理接口, 有时也包括微控制器用于控制DDR的状态”。以上说法是正确的吗? 正确。这里的微控制器主要用于配合整个DDR子系统工作。
- 课后问题4: DDR Controller与DDR PHY之间的接口为DFI接口?
- 课后问题5: DDR的主要功能目标一般包括以下哪些? (不定项选择) A. 将系统命令转换为DDR命令; B. 在系统的统一控制下进入低功耗状态; C. 向系统反馈DDR的中断与状态; D. 对图像进行处理; E. 进行卷积运算; F. 执行安全权鉴, 防止非安全访问。A, B, C, F。这里提到的D属于ISP的工作; E属于NPU的工作。

课后问题

- 课后问题1: DDR Controller的频率为800MHz, DDR PHY的数据速率为6400Mbps。这种情况下, wdata/rdata需要___个phase?
- 课后问题2: 为了保证seamless的data访问, 调度的策略应当是以下哪种? A. 跨BG; B. 相同BG内;
- 课后问题3: 为了保证调度的公平性, 某个一直没有被调度出去的命令, 即使需要打断其他连续seamless的访问, 也应当被调度出去吗? 为什么会出现这种情况?
- 课后问题4: 为了保证调度的效率, 是否应当尽可能经常性地地进行读写切换?
- 课后问题5: 上游地址连续对于DDR的访问效率是否友好?
- 课后问题6: read data channel如果被反压, 是否会影响到ddr的访问效率?
- 课后问题7: open page policy是否优于closed page policy?

下节预告

第八节 - DDR PHY

- DDR PHY的主要功能目标?
- 典型常见特性?
- 主要架构划分?

谢谢